ChatGPT

Kesher Attraction Learning From Past Swipes Dossier

Executive answer

VERIFIED / INFERRED. Past swipes, likes, passes, saves, dwell, and "more like this / less like this" signals can credibly support a **private, provisional taste-calibration system**: a model that learns which visible profile features and presentation patterns a user tends to respond to **on this app, on this surface, at this moment**. They do **not** credibly support claims that the system can predict **dyad-specific chemistry**, mutual compatibility after meeting, or long-run relationship success. That boundary is the central scientific and trust boundary for Kesher. Implicit-feedback systems are useful, but they do not directly observe dislike, and early romantic attraction remains weakly predictable once you try to predict the unique pull between two specific people rather than broad tendencies. [1]

VERIFIED. The strongest external evidence points in one direction: profile-based choice is a **first-impression** phenomenon, often heavily shaped by profile photo attractiveness, presentation order, and curated self-presentation. A 2025 conjoint study of swiping decisions found physical attractiveness dominated other profile attributes in short-term online selection, and experimental work in an online-dating-like paradigm found that attractiveness judgments are sequentially biased by the face seen immediately before the current one. That means swipe logs are real signals, but they are not pure readouts of deep compatibility. [2]

INFERRED. The safest internal framing is **taste calibration** or **private taste profile**. The safest user-facing framing is: **"We use your activity to tune your recommendations. It stays private. You can edit or reset it anytime."** That framing is safer than "attraction learning," "chemistry prediction," or "compatibility scoring" because it is bounded, revisable, and does not imply mind-reading or fate-reading. It also fits the posture already present in the selected repository, which emphasizes "private taste," edit/reset controls, and red lines against attractiveness scoring and sensitive photo inference. [3] [4] [5]

TRADEOFF. Humble framing will be less marketable than hypey claims like "chemistry-first" or "AI compatibility," but it is materially safer for trust, scientific honesty, and consumer-protection risk. The current repository is already close to the right posture in several places, but one visible exception is the Explore surface's "chemistry" mode, which overstates what swipe-derived signals can support and should be renamed or removed. [6] [4] [7]

UNCERTAINTY. Most of the best evidence comes from general-population online dating, speed-dating, and recommender-systems research rather than a serious Jewish dating niche. Kesher's population may place relatively more weight on explicit intent, observance, and values than mass-market swipe apps do. That could raise the usefulness of **explicit** filters and user-stated values, but it does not remove the basic scientific limit on predicting unique chemistry from pre-meeting traces alone. That must be tested locally before any stronger claim is made. [8]

Scientific framing

Scientific term resolution

VERIFIED. In recommender-systems language, swipes and similar behaviors are **implicit feedback**. That means the system learns from actions rather than explicit ratings. The upside is scale and realism; the downside is ambiguity. Implicit data does not directly tell you what a user truly dislikes, and unobserved items are not clean negatives. That is why recommender papers treat these signals as preference evidence with varying confidence rather than as truth. [9]

VERIFIED / INFERRED. In relationship science, the relevant validated constructs are things like **initial attraction**, **romantic desire**, and partner evaluation. “Chemistry” is common in product language, but scientifically it is underspecified and easy to overclaim. The strongest studies show that broad trait- and ideal-based matching has at best modest predictive value, while the unique attraction one specific person feels for one other specific person before they meet is hard to predict from self-reports and pre-date measures alone. [10]

RECOMMENDATION — INFERRED. Internally, Kesher should use **interaction-based taste calibration** as the model name and keep **Private Taste Profile** as the user-facing object. “Preference learning” is acceptable but colder and more profiling-sounding. “Attraction learning” feels creepier and implies something deeper than the evidence supports. “Chemistry prediction” should be a red-line term. This recommendation is trust-safe because it names the capability at the right level of abstraction: a revisable tuning layer, not an identity verdict. [3] [4] [11]

What swipes can and cannot support

VERIFIED. Swipes can support learning of **broad, surface-level patterns** such as: which combinations of explicit intent, age range, distance, observance preferences, verified status, photo style, bio density, prompt style, or profile format a user tends to select or save more often. They can also support relative ranking among visible candidates and can improve local recommendation efficiency, especially when paired with explicit filters and user corrections. [12]

VERIFIED. Swipes cannot by themselves support credible claims about:
a) whether two people will have chemistry in person;
b) whether they are highly compatible as a pair;
c) whether the relationship will succeed; or
d) whether the user’s behavior reveals a stable “type” in any deep or essential sense. The best-known review of online dating concluded that there was no compelling evidence for matching algorithms’ superiority claims, and later machine-learning work found that pre-date traits and preferences failed to predict relationship-specific desire above broad person-level tendencies. Even in richer longitudinal relationship datasets, change in relationship quality was largely unpredictable. [13]

VERIFIED. Swipes are also partly shaped by what the system itself chose to show. Order effects, exposure effects, recent comparisons, and rapid visual adaptation all contaminate behavioral traces. In an online-dating-like experiment, a face was more likely to be judged attractive if the immediately preceding face had been attractive. That means swipes reflect both user taste **and** system presentation. Kesher should therefore treat swipe-derived patterns as **hints under uncertainty**, not ground truth. [14]

Findings and risks

Core findings

VERIFIED. People's abstract ideals and their real-time attraction often diverge. In a classic speed-dating study, participants' ideal preferences assessed before the event failed to predict what inspired their actual desire there. That is highly relevant for Kesher: a model can learn patterns from behavior, but it should not assume those patterns are stable self-knowledge or destiny. [15]

VERIFIED. A newer, very large cross-cultural test complicates the picture in an important way: ideal-partner matching is **not zero**, but it is modest. Across 43 countries and 22 languages, the most rigorous analyses found significant preference-matching effects, but the distinctive effect shrank to a small effect size once normative desirability was modeled alongside it. The right conclusion is not "preferences are useless"; it is "preferences are real but limited, and easy to overstate." [16]

VERIFIED. Short-term swipe choices are especially dominated by first-impression cues. A 2025 conjoint analysis of 5,340 swiping decisions found an overwhelming importance of physical attractiveness relative to bio, intelligence, job, and homophily. For product design, that means a swipe log will overrepresent **first-glance profile appeal** and underrepresent qualities that emerge only later in conversation or in person. Trust-safe recommendation design should acknowledge this rather than pretending the model has learned whole-person compatibility. [17]

VERIFIED. Profiles are curated and sometimes strategically inaccurate. Work on online dating self-presentation has found that daters can successfully present desired personas, and older work found strategic deception especially around physical self-presentation. So, a swipe is partly a signal about the **profile artifact** a user saw, not only the person behind it. This is another reason not to overclaim the depth of what swipes reveal. [18]

VERIFIED. The selected repository already encodes several healthy trust instincts: the AI Trust Hub says AI is an assistant rather than an authority; it explicitly rejects attractiveness ratings and sensitive identity inference from photos; the Private Taste Profile is framed as strictly private, editable, and resettable; and feature metadata marks taste-profile learning as consent-requiring. That is directionally right. The main scientific mismatch is the Explore surface's visible "chemistry" mode. [4] [3] [5] [6]

Fault lines between science and product hype

VERIFIED. The biggest scientific fault line is this: **broad preference learning** is plausible; **dyad-specific chemistry prediction** is not. If product copy collapses those two, the app will sound smarter than the evidence permits. That is exactly the category of unsupported performance claim that regulators scrutinize, and it is the kind of overstatement relationship science has repeatedly warned against in online matching. [19]

VERIFIED. There is also a trust-design fault line between explanation and persuasion. Explanations in recommender systems can serve transparency, scrutability, trust, and persuasiveness, but those aims can conflict. In one influential review, personalization sometimes improved satisfaction while reducing effectiveness. So Kesher should optimize explanations for **truthfulness and user control**, not just for "feeling smart." [20]

VERIFIED. There is a privacy and dignity fault line around inferred sensitive traits. The Information Commissioner's Office [21] states that if a system intends to infer or treat someone differently based on religion, ethnicity, political beliefs, health, sexual orientation, or related categories, that is special-category processing even if the inference is uncertain. The European Data Protection Board [22] likewise treats religion and sexual orientation as sensitive data. In Kesher's context, observance may be user-stated and central to the product, but **hidden inference** from photos or behavior is much riskier than explicit self-description or user-set filters. [23]

VERIFIED. There is a substantiation fault line around AI marketing claims. The Federal Trade Commission [24] has repeatedly emphasized that there is "no AI exemption" from deception law and has challenged unsupported claims about AI systems' performance and bias. The National Institute of Standards and Technology [25] and ICO both emphasize transparency, accountability, context, and impact in trustworthy AI design. In practice, that means Kesher should only claim what it can test and evidence. [26]

What Kesher must never imply

RED LINE — VERIFIED / INFERRED. Kesher must never imply that it can **predict chemistry**, **know who you will fall for in person**, or **forecast relationship success** from swipe history. That goes beyond the evidence and is exactly where science and product hype diverge. [13]

RED LINE — VERIFIED. Kesher must never imply it has inferred a user's religion, ethnicity, sexual orientation, attractiveness, or other sensitive or dignity-laden characteristics from photos or passive behavior. That is both trust-dangerous and legally risky. [23]

RED LINE — INFERRED. Kesher must never imply that a pass means the user "dislikes" a kind of person in any deep sense. A pass can reflect mood, order effects, photo quality, timing, attention, or the simple fact that the profile was shown in the wrong context. That copy would feel accusatory and brittle. [27]

RED LINE — VERIFIED / INFERRED. Kesher must never imply that the other person was exposed because the system knows they are "perfect for you" or "very likely to like you back," unless it has direct, user-consented evidence for that specific claim. Otherwise it is inventing certainty about another person's private preferences. [28]

Product design

Safe product framings

RECOMMENDATION — INFERRED. Keep the core capability framed as **Private Taste Profile** externally and **taste calibration** internally. This is trust-safe because it is accurate, bounded, editable, and already consistent with the repository's privacy-and-control posture. Users are more likely to feel respected when the system says it is **tuning recommendations**, not defining identity. [3] [4]

RECOMMENDATION — HEURISTIC. On **onboarding**, introduce the feature as optional and reversible: "If you want, we can use your activity to tune your recommendations over time." Add three trust anchors in the same breath: **private**, **editable**, **resettable**. This is trust-safe because it reduces creepiness, increases scrutability, and prevents the feeling that the app is silently profiling the user. [11] [3]

RECOMMENDATION — HEURISTIC. On **Daily Picks**, use strongest learning only where the app is already making a finite, intentional curation claim. The repository's existing "finite and intentional" tone is good, and it is directionally aligned with evidence that huge pools can push users into an evaluative, objectifying mindset. Trust-safe copy here is: "These picks are tuned from your goals, filters, and recent activity." That is better than "best matches" or "most compatible." [29] [30]

RECOMMENDATION — HEURISTIC. On **Explore**, de-emphasize learned certainty and emphasize browsing control. The current "chemistry" mode is unsafe because it implies a deeper claim than the model can substantiate. Replace it with something like **Broaden my usual picks**, **Serendipity**, or **Explore beyond my pattern**. That is trust-safe because it honestly describes exploration as exploration. [6] [31]

RECOMMENDATION — INFERRED. On **explanation cards**, constrain reasons to **whitelisted, user-visible, and contestable** signals: shared intent, overlapping explicit values or observance preferences, distance, activity recency, and the user's recent explicit feedback patterns. Never explain with hidden latent claims such as "you're drawn to emotionally unavailable people" or "you prefer X type." That is trust-safe because users can inspect and correct the basis of the explanation. [29] [32] [11]

Unsafe framings and red-line copy

UNSAFE — INFERRED. "We know your type."
Why unsafe: it essentializes the user, sounds judgmental, and overstates stability. It turns a provisional model into an identity claim.

UNSAFE — VERIFIED / INFERRED. "Chemistry-first."
Why unsafe: "chemistry" implies relationship-specific attraction that pre-meeting data cannot reliably predict. The term encourages users to overread the system's confidence. [31]

UNSAFE — VERIFIED. "Predicted compatibility" or "high compatibility score."
Why unsafe: the evidence base for superior algorithmic matching in romance remains weak, and regulators care about unsupported performance claims. [19]

UNSAFE — VERIFIED. "We learned you prefer religious / secular / urban / artistic people" if that conclusion is inferred from passive behavior rather than set explicitly by the user.
Why unsafe: it can cross into sensitive-trait inference and can make users feel categorized or exposed. [33]

UNSAFE — INFERRED. "They're likely to like you back."
Why unsafe: it implies private inference about the other person and can mislead the user about reciprocity and expectations.

Validation plan

Verification experiments

HEURISTIC. Run an **offline calibration study** first. Ask only one question: does the taste model improve held-out ranking accuracy for near-term actions such as like, save, or "more like this," compared with explicit-filter-only baselines? Do not treat this as evidence of chemistry or compatibility. This experiment is trust-safe because it tests the exact bounded claim you want to make. [34]

HEURISTIC. Run an **explanation honesty experiment**. Compare three conditions: no explanation, concrete bounded explanation, and stronger hype-like explanation. Measure not only engagement but also "felt understood," "felt judged," "felt creepy," and "felt accurate." Use human review to ensure explanations are faithful to the actual ranking logic. This is trust-safe because explanation systems can trade off trust, persuasiveness, and effectiveness. [35]

HEURISTIC. Run a **scrutability experiment**. Compare opaque learning versus a visible Private Taste Profile that users can edit, reset, and steer. Success metrics should include comprehension, reset confidence, opt-in rate, and long-term satisfaction, not just clicks. This is trust-safe because user models become less creepy when they are reviewable and correctable. [11]

HEURISTIC. Run an **overfitting-versus-serendipity experiment**. Compare a pure exploitation ranker with a ranker that deliberately broadens a small share of impressions. Watch for diversity, reply quality, save rate, hide rate, block rate, and recommendation regret. This is trust-safe because excessive narrowing can make users feel boxed in by their past behavior. [36]

HEURISTIC / VERIFIED. Run a **sensitive-inference audit** before launch. Explicitly test whether profile images, language style, or behavior proxies allow the model to infer or operationalize religion, ethnicity, sexual orientation, or similar categories beyond what users explicitly disclosed or filtered for. This is trust-safe because these are precisely the categories regulators treat as sensitive when inferred and used for differential treatment. [23]

UNKNOWN. If Kesher ever wants to say anything stronger than "tunes your picks," it needs a **prospective, user-consented longitudinal study** with downstream outcomes beyond swipes, such as conversation continuity, perceived explanation accuracy, post-date mutual interest, and later relationship continuation. Until then, stronger claims should remain off-limits.

UX copy and recommendation

Recommended UX language

Onboarding — HEURISTIC.
"Optional: use my activity to tune my recommendations."
"Kesher can learn from likes, passes, saves, and 'more like this' to improve your picks."
"This stays private, is never shown to other members, and you can edit or reset it anytime."
Why this is safe: it is explicit, bounded, private, and reversible. [3] [37]

Daily Picks — HEURISTIC.
"Today's picks are tuned from your goals, filters, and recent activity."
"Finite, intentional, and adjustable."
Why this is safe: it claims curation, not certainty, and fits the existing repository tone. [29]

Explore — HEURISTIC.
Replace “Chemistry” with “Broaden my usual picks.”
Secondary helper text: “See a wider mix than your recent pattern.”
Why this is safe: it describes exploration honestly and avoids unsupported prediction language. [6]

Explanation card — HEURISTIC.
“You’re seeing this because you both want a serious relationship, share similar observance preferences, and you’ve recently engaged more with profiles like this.”
Then offer two user actions: **More like this** and **Less like this**.
Why this is safe: it uses visible, contestable reasons and immediately gives the user steering control. [29]

Private Taste Profile page — HEURISTIC.
“These are provisional patterns from your activity, not fixed labels.”
“Update them, pause them, or reset them anytime.”
Why this is safe: it lowers identity threat and keeps the model scrutable.

Bottom-line recommendation

FINAL RECOMMENDATION — VERIFIED / INFERRED. Ship this as a **private, editable recommendation-tuning system**, not as chemistry prediction and not as compatibility scoring. Internally, call it **taste calibration**. Externally, keep **Private Taste Profile** and make the public claim no stronger than: **“We use your activity to tune your picks.”** Use explanations only for concrete, whitelisted, user-visible reasons; keep controls like edit/reset/more-like-this/less-like-this; and remove or rename any “chemistry” language from Explore and related surfaces. That is the highest-confidence product-safe, trust-safe, dignity-preserving position supported by the current evidence. [38] [3] [6] [4]

Open questions and limitations

UNKNOWN / INFERRED. The main open question is not whether taste calibration can improve near-term recommendation quality; it likely can. The open question is how much value explicit intent, observance, and serious-dating norms add in Kesher’s specific population beyond general swipe-style evidence. That should be answered with Kesher-specific experiments before any broader marketing claim is made.

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